CHEM 427/527 · SESSION 5 · THURSDAY, OCTOBER 15, 2026

Tidy data and the sample table

Matt Bush, with teaching assistants Lucas Narisawa and Chris Weir

Drafted with assistance by Claude Code. Reviewed and revised by Matt.

LEARNING OBJECTIVES

By the end of today you can

1. State the three rules of tidy data, and point to the cell, column, or row in a wide laboratory report that breaks each one.
2. Reshape a wide results table to one measurement per row with pandas, split a composite sample identifier into its parts, and encode a below-reporting-limit result as missing rather than zero.
3. Join the measurement table to the sample table on its key with a left merge, and say what an inner merge would have discarded.
4. Compute a detection rate and a mean detected concentration by group, and say how the choice of what counts as an observation changes the number.

FIRST, EVERYONE PULLS

Get session 5 with git pull

Bring in session 5

```
git pull
```

A new folder appears, with 114 files under data/states/:

`S05-tidy-data`

Session 4 is due today

- Upload it by 1:30 pm.
- Still no working install? The deadline was Tuesday. See Matt in the third hour.

Nothing to download today: every state's two files are in the repository.

TODAY'S ANALYTE

Why water is tested for PFAS

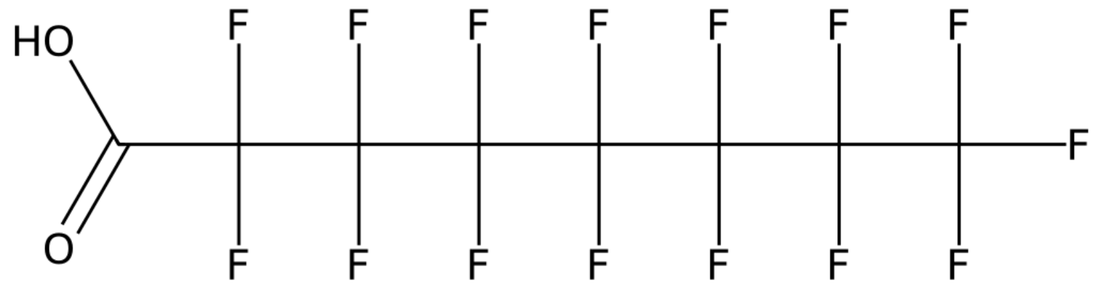


Figure 1. PFOA, perfluorooctanoic acid.

PFAS, per- and polyfluoroalkyl substances: several thousand synthetic compounds on a carbon chain whose hydrogens are replaced by fluorine.

Made since the 1940s for nonstick coatings, stain-resistant fabrics, and firefighting foams.

The carbon-fluorine bond breaks down very slowly, so PFAS accumulate in soil and water.

EPA: exposure to some PFAS may be linked to harmful health effects. Limits for six PFAS were set in April 2024; a May 2026 proposal keeps those for PFOA and PFOS.

Sources: EPA, PFAS Explained and PFAS drinking water regulation; PubChem record for PFOA.

WHERE SESSION 4 STOPPED

A row is one measurement

Tuesday

1 spectrum

88 lines, ten points a line, and a header that said how to read them.

Today, EPA's UCMR 5 release

1,992,002

PFAS measurements in drinking water, from 57 states and territories and 9 EPA Region tribal programs, one measurement per row.

What would a row have to hold for you to compute a PFOA detection rate by state and by water type in one line?

WHO ASKS THE QUESTION

A water system decides on treatment

4.0 ng/L

EPA's 2024 standard for PFOA in drinking water, averaged over a year.

0.004 µg/L

The same number in today's files, which report µg/L. It is also the reporting limit.

Plan treatment for water that meets the standard, and the customers pay for a carbon plant they did not need. Skip it for water that does not, and the same customers drink water above the standard.

TIDY DATA: ONE OBSERVATION PER ROW, ONE VARIABLE PER COLUMN, ONE VALUE PER CELL

Easy to read, hard to compute on

The first three rows of WA_results.csv. **The PFAS laboratory report breaks all three rules of tidy data.**

sample_id	PFBS	PFHxA	PFHxS	PFOA	PFOS	PFPeA
WA5300050_000509_2024-01-17_SE1	< 0.003	< 0.003	< 0.003	< 0.004	< 0.004	< 0.003
WA5300100_EP1_2024-02-27_SE1	< 0.003	< 0.003	< 0.003	< 0.004	< 0.004	< 0.003
WA5300431_EP001_2024-05-20_SE1	< 0.003	< 0.003	< 0.003	< 0.004	< 0.004	< 0.003

One variable per column

PFOA is a column name. The compound is a variable, and needs a column of its own.

One observation per row

A row holds six measurements, one for each compound.

One value per cell

The sample_id joins four facts: system, sample point, date, event. **< 0.004** is a code: below the reporting limit.

FOUR STEPS TO TIDIER DATA: TWO OF THEM LIVE, ON THE THREE-ROW STUB

Split one column, melt six

```
stub["sample_id"].str.split("_", expand=True)
```

0	1	2	3
WA5300050	000509	2024-01-17	SE1
WA5300100	EP1	2024-02-27	SE1
WA5300431	EP001	2024-05-20	SE1

Four columns from one: a value per cell.

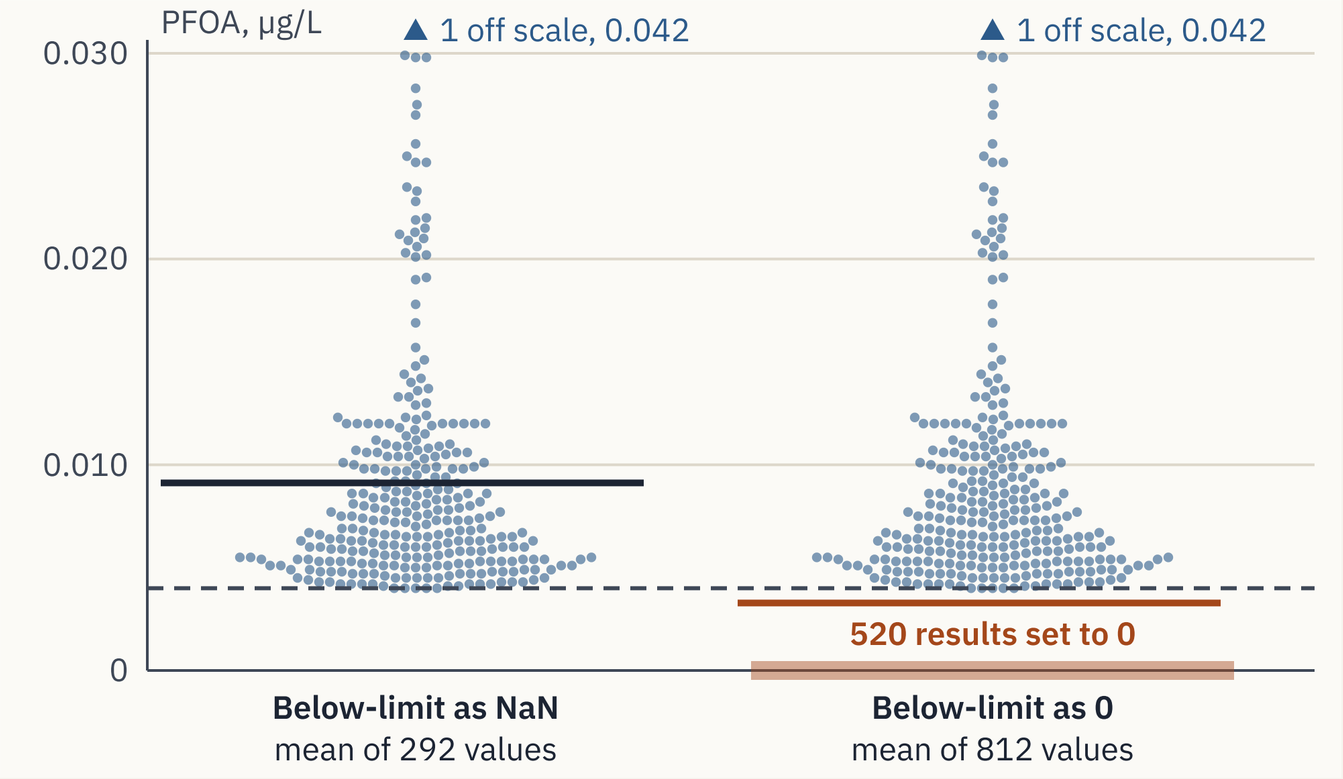
```
stub.melt(id_vars=["sample_id"], var_name="compound", value_name="concentration")
```

	sample_id	compound	concentration
0	WA5300050_000509_2024-01-17_SE1	PFBS	< 0.003
…	…	…	…
9	WA5300050_000509_2024-01-17_SE1	PFOA	< 0.004
10	WA5300100_EP1_2024-02-27_SE1	PFOA	< 0.004
11	WA5300431_EP001_2024-05-20_SE1	PFOA	< 0.004
17	WA5300431_EP001_2024-05-20_SE1	PFPeA	< 0.003

Eighteen rows from three, and PFOA is now a value you can filter on.

BELOW THE REPORTING LIMIT IS NOT ZERO

Zeros move the mean below the MCL



New Jersey, first sampling event: 812 PFOA results, 292 detected, 520 reported as **< 0.004**.

0.00912, the mean of what was measured

0.00328, the mean with 520 zeros: 292 × 0.00912 / 812

0.004, the MCL, which every detection meets or exceeds

A below-limit result is a measurement that was made, not a zero. Store it as NaN.

WHAT ONE ROW STANDS FOR: NEW JERSEY, PFOA

Same detections, three denominators

Per result

36.0%

292 of 812 results

Per sample point

36.0%

292 of 812 sample points

Per system

50.0%

131 of 262 systems

```
my_pfoa["detected"].mean()
my_pfoa.groupby(["pwsid", "sample_point_id"])["detected"].any().mean()
my_pfoa.groupby("pwsid")["detected"].any().mean()
```

Section 3 of the notebook computes all three for your state.

ASSIGNMENTS

One state each

#	Code	State	Systems
1	TX	Texas	1143
2	CA	California	700
3	IL	Illinois	487
4	FL	Florida	400
5	PA	Pennsylvania	389
6	OH	Ohio	346
7	NY	New York	332
8	MI	Michigan	319
9	AL	Alabama	306

#	Code	State	Systems
10	NC	North Carolina	285
11	TN	Tennessee	277
12	MA	Massachusetts	263
13	GA	Georgia	254
14	MS	Mississippi	236
15	MO	Missouri	232
16	KY	Kentucky	231
17	IN	Indiana	230
18	WI	Wisconsin	202

#	Code	State	Systems
19	MN	Minnesota	196
20	VA	Virginia	171
21	CO	Colorado	164
22	SC	South Carolina	161
23	AZ	Arizona	151
24	OR	Oregon	127
25	WV	West Virginia	102
26	MD	Maryland	83

Set **STATE** in Parameters to your two-letter code. Your two files are already in data/states/.

AI PRACTICE 2 OF 8: SPECIFIC PROMPTS

Name the table, the rule, and the check

Short: gets code that runs

My PFAS columns hold strings like "< 0.004". How do I make them numeric?

Specific, before TASK 2: gets code you can check

In my notebook, my_results is **data/states/WA_results.csv**. Its **six PFAS columns** hold numbers as text, and strings that start with "<" for a result below the reporting limit. Show me how to make those columns numeric so that **a below-limit result becomes NaN**, and how to check that **the number of NaN values in each column equals the number of "<" strings before**.

In bold: the table, the columns, how to store a result below the reporting limit, and a check that the conversion lost nothing.

Which of the two answers could you check against Washington's table?

THE JUDGMENT CELL, SECTION 4

Your rates, and one entry

(a) Your answer

- Your PFOA detection rate per result and per system, each with its denominator
- The mean detected PFOA concentration
- Which rate answers the systems question, and why
- What the mean would have been with the below-limit results as zero

(b) One judgment entry

- What the assistant suggested
- What you did: changed, rejected, or verified
- Why, from a count or a value in your state's tables

Scored 0 to 2. The two prompts in section 2 are the sources; the specific one is the expected one.

SECTION 1 IS READ AND RUN. SECTION 2 IS YOURS.

Go, then submit

Submit

Restart, Run All, save, and upload activity.ipynb by **1:30 pm Tuesday, October 20**.

Every state's files are in the repository, so the graders' run reads the same data yours did.

Scoring, 0 to 3

1 if it runs after a restart, **0** if not.

0 to 2 for the judgment cell.

Help

Ask Matt, Lucas, or Chris in the room.

The third hour is an office hour.